Week 4 Discussion

Problem 1 (High-water marks: from a cdf to a pdf, and a change of units; C&B Exercise 1.53)

A certain river floods every year. Suppose that the low-water mark is set at 1 and the high-water mark Y has distribution function

$$F_Y(y) = P(Y \le y) = 1 - \frac{1}{y^2}, \qquad 1 \le y < \infty,$$

and $F_Y(y) = 0$ for $y < 1$.

(a) Verify that $F_Y(y)$ is a cdf, by checking the three conditions of Theorem 1.5.3. Is Y continuous, discrete, or neither?

(b) Find $f_Y(y)$, the pdf of Y. Find $P(Y > 2)$ twice: once from F_Y, and once by integrating f_Y.

(c) If the low-water mark is reset at 0 and we use a unit of measurement that is $\frac{1}{10}$ of that given previously, the high-water mark becomes $Z = 10(Y - 1)$. Find $F_Z(z)$. *Hint:* write the event $\{Z \le z\}$ as an event about Y.

(d) Find $f_Z(z)$ in two ways: by differentiating your answer to (c), and by Theorem 2.1.5. What are $\mathcal{Y}$ and $\mathcal{Z}$?

▶ *Show answer*

Problem 2 (A discrete transformation; C&B Exercise 2.3)

Suppose X has the geometric pmf $f_X(x) = \frac{1}{3}\left(\frac{2}{3}\right)^x$, $x = 0, 1, 2, \ldots$.

(a) Verify that f_X is a pmf (Theorem 1.6.5). Then find the cdf $F_X(x)$ for **every** real x. *Hint:* use the partial sum of the geometric series, as in (1.5.4).

(b) Determine the probability distribution of $Y = X/(X+1)$. Note that here both X and Y are discrete random variables. To specify the probability distribution of Y, specify its pmf.

- What is $\mathcal{Y}$?
- Is $g(x) = x/(x+1)$ one-to-one on $\mathcal{X}$? What is $g^{-1}(y)$?

(c) Find $P(Y \le 0.7)$.

(d) All the values of Y lie in $[0, 1)$, and they pile up near 1. Is Y still a discrete random variable? Explain using Definition 1.5.7.

▶ *Show answer*

Problem 3 (When the pieces don't match: a piecewise-monotone transformation; C&B Exercise 2.7)

Let X have pdf $f_X(x) = \frac{2}{9}(x+1)$, $-1 \le x \le 2$, and let $Y = X^2$.

(a) Sketch f_X and the graph of $g(x) = x^2$ on $[-1, 2]$. What is $\mathcal{Y}$? Split $\mathcal{X}$ into the two pieces $(-1, 0)$ and $(0, 2)$ on which g is monotone, and find the image of each piece. Which condition of Theorem 2.1.8 fails? (This is why C&B note that Theorem 2.1.8 is not directly applicable in this problem.)

(b) Use the cdf method to find $F_Y(y)$. *Hint:* the event $\{X^2 \le y\} = \{-\sqrt{y} \le X \le \sqrt{y}\}$ looks different depending on whether $-\sqrt{y}$ is inside the support of X. Treat $0 \le y < 1$ and $1 \le y \le 4$ separately.

(c) Differentiate to find the pdf of Y. Check that it integrates to 1. Is it bounded?

(d) Show that Theorem 2.1.8 remains valid if the sets $A_0, A_1, \ldots, A_k$ *contain* $\mathcal{X}$, and apply the extension to solve part (c) using $A_0 = \emptyset$, $A_1 = (-2, 0)$, and $A_2 = (0, 2)$. *Hint:* outside the support, $f_X = 0$.

▶ *Show answer*